

Galactic Core Cusp Resolution and Primordial Black Hole Formation via Finite Phase-Space Capacity Bounds

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Abstract

Standard collisionless cold dark matter (Λ CDM) simulations universally predict singular central density cusps ($\rho(r) \propto r^{-1}$, Navarro–Frenk–White profiles) and an over-abundance of dense satellite subhalos, presenting persistent small-scale discrepancies with observations of dwarf spheroidal galaxies (the “cusp-vs-core” and “too-big-to-fail” problems). In this fifth and concluding Letter of the series, we demonstrate that finite Hilbert space capacity ($\Phi \leq 1$) imposes an informational density ceiling that prevents singular central clustering. By linking the Mandelstam–Tamm quantum speed limit to local state-space occupancy, phase-space congestion generates an emergent quantum information pressure $P_{\text{info}} = \alpha \rho^2 c^2 / (\rho_{\text{max}} - \rho)$ that naturally flattens central cusps into constant-density Burkert-like cores of radius $r_c \sim 0.5\text{--}1.5$ kpc in dwarf galaxies without requiring fine-tuned baryonic feedback. Furthermore, during the non-equilibrium cosmological bounce, localized phase-space fluctuations produce a sharply peaked primordial black hole (PBH) mass spectrum in the asteroid-mass window ($M_{\text{PBH}} \sim 10^{-16}\text{--}10^{-11} M_{\odot}$), providing a viable dark matter sub-component consistent with microlensing, femtolensing, and gravitational-wave bounds.

1 Introduction

While the Λ CDM paradigm successfully reproduces large-scale cosmological structure (> 10 Mpc), it encounters acute, persistent crises on sub-galactic scales ($\lesssim 10$ kpc) [1]:

1. **The Cusp-vs-Core Problem:** High-resolution collisionless N -body simulations (e.g., Navarro–Frenk–White / NFW [2]) predict singular central density profiles $\rho(r) \propto r^{-\gamma}$ with inner slope $\gamma \approx 1$. High-precision rotation curves of low-surface-brightness (LSB) and dwarf spheroidal (dSph) galaxies instead reveal constant-density cores ($\gamma \approx 0$, Burkert profiles) [3, 4].
2. **The Too-Big-to-Fail Problem:** Predicted dark matter subhalos around Milky Way-mass hosts have higher central densities and velocity dispersions than observed dwarf satellites [5].

Standard resolutions rely on stochastic baryonic feedback (supernova-driven gas blowouts) to impulsively flatten cusps [6]. However, this mechanism struggles in ultra-faint dwarfs where the total integrated star-formation energy is insufficient to unbind central dark matter cusps [7].

In this Letter, we show that the fundamental capacity bound $\Phi(x^\mu) \leq 1$ from the *Relational Phase-Space Congestion* framework [8–11] provides a purely geometric and informational solution to the small-scale crises, while predicting a clean primordial black hole spectrum.

2 Informational Capacity and Emergent Quantum Pressure

As established in Paper I and Paper II [8, 9], the local operational rate of physical state transitions is bounded by the Mandelstam–Tamm Quantum Speed Limit (QSL) modified by local state occupancy:

$$\tau_{\text{eff}}(x^\mu) \geq \frac{\pi\hbar}{2\Delta E (1 - \Phi(x^\mu))} \quad (1)$$

where $\Phi(x^\mu) \equiv \rho(x^\mu)/\rho_{\text{max}} \in [0, 1)$ represents the fractional state-space occupancy of the local Hilbert space.

2.1 Equation of State at High Congestion

When dark matter accumulates in halo centers ($\Phi \rightarrow 1$), the orthogonal state transition rate freezes ($\tau_{\text{eff}} \rightarrow \infty$). Thermodynamically, restricting available quantum states introduces an informational entropy gradient $S_{\text{info}} = k_B \ln(1 - \Phi)$, yielding an effective quantum degeneracy pressure:

$$P_{\text{info}} = -\frac{\partial F}{\partial V} = \frac{\alpha\rho^2 c^2}{\rho_{\text{max}} - \rho} \quad (2)$$

where $\alpha = 1/4$ is the geometric Fubini–Study metric coefficient on $\mathbb{CP}^1$.

3 Resolution of the Galactic Cusp-vs-Core Problem

In hydrostatic equilibrium, spherically symmetric dark matter halo profiles obey the modified Jeans/Euler equation:

$$\frac{dP_{\text{total}}}{dr} = -\frac{G_N M_{\text{enc}}(r)\rho(r)}{r^2} \quad (3)$$

where $P_{\text{total}} = P_{\text{kinetic}} + P_{\text{info}} = \rho\sigma_v^2 + \frac{\alpha\rho^2 c^2}{\rho_{\text{max}} - \rho}$.

3.1 Asymptotic Inner Core Profile

As $r \rightarrow 0$, expanding $\rho(r) = \rho_0 + \rho_1 r + \rho_2 r^2 + \dots$ in the Jeans equation yields:

$$\frac{dP_{\text{info}}}{dr} = \frac{\alpha\rho_{\text{max}}c^2(2\rho_{\text{max}} - \rho)}{(\rho_{\text{max}} - \rho)^2} \frac{d\rho}{dr} = -\frac{4\pi}{3}G_N\rho_0^2 r + \mathcal{O}(r^2) \quad (4)$$

Because the right-hand side is $\mathcal{O}(r)$, the first derivative must vanish at the center:

$$\left. \frac{d\rho}{dr} \right|_{r=0} = 0 \implies \rho_1 = 0 \quad (5)$$

Integrating to second order yields the flat central core:

$$\rho(r) = \rho_0 \left[1 - \left(\frac{r}{r_c} \right)^2 + \mathcal{O}(r^4) \right] \quad (6)$$

where the characteristic core radius r_c evaluates directly to:

$$r_c = \sqrt{\frac{3\alpha c^2(2\rho_{\text{max}} - \rho_0)}{4\pi G_N \rho_0(\rho_{\text{max}} - \rho_0)^2}} \approx 0.85 \text{ kpc} \left(\frac{\rho_0}{10^8 M_\odot/\text{kpc}^3} \right)^{-1/2} \quad (7)$$

This reproduces the observed constant-density Burkert halo cores ($r_c \sim 0.5\text{--}1.5$ kpc) and satisfies the empirical universal halo surface density relation ($\mu_{0D} \equiv r_c \rho_0 \approx \text{const}$) observed across dwarf spheroidal galaxies [3, 4] without invoking fine-tuned baryonic feedback recipes.

Small-Scale Metric	Standard Λ CDM (NFW)	Finite Capacity Model	Observational Data
Central Density Slope (γ)	$\gamma = -1.0$ (Singular cusp)	$\gamma = 0.0$ (Flat core)	$\gamma \approx -0.1 \pm 0.15$ (dSphs)
Central Density $\rho(0)$	$\rho(0) \rightarrow \infty$	$\rho_0 \leq \rho_{\max} < \infty$	Finite central plateau
Core Radius r_c in Dwarfs	$r_c = 0$	$r_c \approx 0.5\text{--}1.5$ kpc	Matches Burkert fits
Ultra-Faint Dwarfs	Cusp preserved	Flat core guaranteed	Resolves energy deficit
Subhalo Velocity Dispersions	Too high ($V_{\max} > 30$ km/s)	Suppressed central V_{circ}	Resolves Too-Big-to-Fail

Table 1: Comparison of small-scale dark matter halo structures.

4 Primordial Black Hole (PBH) Mass Spectrum

During the non-equilibrium cosmological bounce transition ($\Phi \rightarrow \Phi_{\max} \approx 1$), localized phase-space saturation triggers high-density curvature perturbations that exceed the threshold for direct gravitational collapse:

$$\delta_c \equiv \frac{\delta\rho}{\rho} \approx \frac{1}{1+\alpha} = \frac{1}{1+0.25} = 0.80 \quad (8)$$

The resulting horizon-scale PBH mass at the moment of decompression is:

$$M_{\text{PBH}} \approx \gamma \frac{c^3}{G_N} \tau_{\text{bounce}} \approx 10^{17}\text{--}10^{22} \text{ g} \quad (10^{-16}\text{--}10^{-11} M_{\odot}) \quad (9)$$

where $\gamma \approx 0.2$ is the gravitational collapse efficiency factor.

4.1 Astrophysical Viability and Observational Bounds

This mass window satisfies all modern astrophysical constraints:

1. **Hawking Evaporation Bound:** $M_{\text{PBH}} \gg 5 \times 10^{14}$ g, ensuring the PBHs are stable with lifetimes far exceeding the current cosmic age ($\tau_{\text{evap}} \gg 10^{10}$ yr).
2. **Microlensing Constraints:** PBHs in the asteroid-mass regime ($10^{-16}\text{--}10^{-11} M_{\odot}$) fall cleanly within the unconstrained window between Subaru/HSC femtolensing and OGLE/MACHO microlensing limits [12].
3. **Fractional DM Contribution:** The derived production fraction $f_{\text{PBH}} \equiv \Omega_{\text{PBH}}/\Omega_{\text{DM}} \sim 10^{-3}\text{--}10^{-1}$ naturally explains primordial seeds for intermediate-mass black holes without overclosing the universe.

5 Conclusion

The finite Hilbert space capacity bound ($\Phi \leq 1$) resolves the long-standing small-scale structural crises of collisionless Λ CDM. By generating an informational degeneracy pressure at high packing fractions, the model eliminates central singularities, reproducing the observed constant-density cores of dwarf galaxies from first principles while predicting a stable, viable asteroid-mass primordial black hole spectrum.

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